COMP 416

Network Layer Concepts Practice and Simulations Using Cisco Packet Tracer

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Part 1. Network Layer Analysis through ICMP

I used the *traceroute* www.mcgill.ca command to perform a traceroute to the URL and find the IP addresses of the routers that forward the message.

1. If UDP ports or ICMP Echo probes are blocked by the destination firewall, explain which traceroute options you would modify to successfully complete the trace, and why. No screenshot is expected (Refer to traceroute documentation or man pages)

As mentioned in the traceroute manual [1], traceroute can be configured to use different probe methods when UDP ports or ICMP Echo probes are blocked by a firewall. In such cases, TCP or UDP probes using a constant destination port can be used to bypass firewall restrictions. Since commonly used ports are more likely to be allowed, the traceroute packets have a higher chance of reaching the destination. This can be achieved by using TCP-based or UDP-based traceroute options. For instance, TCP probes can be used explicitly with this command: *traceroute -P tcp www.mcgill.ca*

2. Find the minimum TTL less than which the traceroute messages do not reach your particular URL destination. Provide a screenshot satisfying your answer.

Traceroute sends probe packets starting with a TTL value of 1 and increases the TTL by one at each hop. When a probe reaches a router, an ICMP Time Exceeded message is returned, and the router's IP address is shown. If no response is received within the timeout period, an asterisk (*) is displayed. It means that the probe did not get a reply [1].

From the traceroute output, replies are received up to hop 13. Starting from hop 14, the output shows * * *, meaning that all three probes timed out and no response was returned. Therefore, the minimum TTL value for which the traceroute messages do not reach the destination is TTL = 14.

```
(base) zulaldemirtas@Zulals-MacBook-Pro-2 ~ % traceroute www.mcgill.ca
traceroute to webcache.it.mcgill.ca (132.216.98.100), 64 hops max, 40 byte packets
 1  172.21.128.2 (172.21.128.2)  5.222 ms  4.874 ms  4.079 ms
 2  10.20.30.2 (10.20.30.2)  4.798 ms  5.597 ms  4.833 ms
 3  212.175.32.141.static.ttnet.com.tr (212.175.32.141)  5.052 ms  4.986 ms  5.111 ms
 4  212.174.167.209 (212.174.167.209)  7.747 ms  7.628 ms  6.816 ms
 5  00-gayrettepe-sr14s-t2-1---00-gayrettepe-guvenlik-t3-1.statik.turktelekom.com.tr (81.212.221.114)  3545.031 ms *  3534.021 ms
 6  * * *
 7  301-fra-col-2---00-ebgp-gayrettepe-sr12e-k.statik.turktelekom.com.tr (212.156.101.196)  39.518 ms  97.275 ms  42.063 ms
 8  * 212.222.126.8 (212.222.126.8)  44.207 ms  43.826 ms
 9  ae16.cr0-mtl1.ip4.gtt.net (89.149.186.134)  135.846 ms  135.180 ms  204.273 ms
10  ip4.gtt.net (208.116.159.206)  142.408 ms  144.053 ms  143.089 ms
11  imtrl-rq-ic-dmtrl-rq.risq.net (192.77.63.70)  155.650 ms  154.408 ms  155.096 ms
12  * * *
13  mcgill-gw.risq.net (132.202.62.90)  154.669 ms  184.924 ms  155.247 ms
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
```

Figure 1: Traceroute output to www.mcgill.ca

3. Observe and report how many hops does it take to leave your local site. Explain your answer. No screenshot is expected.

As can be seen in Figure 1, the first two hops belong to private IP address ranges, 172.21.x.x and 10.x.x.x, which correspond to the local network [2]. The third hop has a public IP address, indicating that the traffic has left the local site. Therefore, it takes three hops to leave the local site.

4. What is the default number of probes used by the traceroute? Run multiple traceroutes, increasing the number of probes progressively. Explain your observation regarding the resolution of the route to your destination IP address. Provide a screenshot satisfying your answer.

According to the manual, the default number of probes used by traceroute is 3 [1]. I ran traceroute four times using different probe values: q = 1, q = 3, q = 5, and q = 10. When the number of probes is set to 1, only one response is collected per hop. This results in an output with lower resolution, since some hops are shown as * due to packet loss or delayed responses. For q = 3, 5, and 10, the resolution improves because lost packets are compensated by additional

probes. However, increasing the number of probes also increases the overall execution time of traceroute, which can be inefficient.

```
(base) zulaldemirtas@Zulals-MacBook-Pro-2 ~ % traceroute -q 1 www.mcgill.ca

traceroute to webcache.it.mcgill.ca (132.216.98.100), 64 hops max, 40 byte packets
 1  172.21.128.2 (172.21.128.2)  6.221 ms
 2  10.20.30.2 (10.20.30.2)  6.566 ms
 3  212.175.32.141.static.ttnet.com.tr (212.175.32.141)  5.140 ms
 4  212.174.167.209 (212.174.167.209)  8.659 ms
 5  00-gayrettepe-sr14s-t2-1---00-gayrettepe-guvenlik-t3-1.statik.turktelekom.com.tr (81.212.221.114)  3519.571 ms
 6  *
 7  301-fra-col-2---00-ebgp-gayrettepe-sr12e-k.statik.turktelekom.com.tr (212.156.101.196)  60.904 ms
 8  *
 9  ae16.cr0-mtl1.ip4.gtt.net (89.149.186.134)  196.433 ms
10  ip4.gtt.net (208.116.159.206)  141.620 ms
11  imtrl-rq-ic-dmtrl-rq.risq.net (192.77.63.70)  156.141 ms
12  *
13  *
```

Figure 2: Traceroute output with probe = 1

```
(base) zulaldemirtas@Zulals-MacBook-Pro-2 ~ % traceroute -q 3 www.mcgill.ca

traceroute to webcache.it.mcgill.ca (132.216.98.100), 64 hops max, 40 byte packets
 1  172.21.128.2 (172.21.128.2)  5.639 ms  5.254 ms  3.666 ms
 2  10.20.30.2 (10.20.30.2)  5.001 ms  6.040 ms  4.079 ms
 3  212.175.32.141.static.ttnet.com.tr (212.175.32.141)  4.999 ms  4.907 ms  5.421 ms
 4  212.174.167.209 (212.174.167.209)  7.910 ms  9.224 ms  6.190 ms
 5  00-gayrettepe-sr14s-t2-1---00-gayrettepe-guvenlik-t3-1.statik.turktelekom.com.tr (81.212.221.114)  3496.239 ms  3489.749 ms  3458.136 ms
 6  00-ebgp-gayrettepe-sr12e-k---00-gayrettepe-sr14s-t2-1.statik.turktelekom.com.tr (81.212.220.238)  3542.838 ms * *
 7  301-fra-col-2---00-ebgp-gayrettepe-sr12e-k.statik.turktelekom.com.tr (212.156.101.196)  43.259 ms  40.527 ms  39.636 ms
 8  * * *
 9  ae16.cr0-mtl1.ip4.gtt.net (89.149.186.134)  151.097 ms  183.731 ms  200.035 ms
10  ip4.gtt.net (208.116.159.206)  141.435 ms  187.930 ms  192.308 ms
11  imtrl-rq-ic-dmtrl-rq.risq.net (192.77.63.70)  166.885 ms  156.712 ms  200.064 ms
12  * * *
13  mcgill-gw.risq.net (132.202.62.90)  156.378 ms  159.180 ms  190.918 ms
14  * * *
```

Figure 3: Traceroute output with probe = 3

```
(base) zulaldemirtas@Zulals-MacBook-Pro-2 ~ % traceroute -q 5 www.mcgill.ca

traceroute to webcache.it.mcgill.ca (132.216.98.100), 64 hops max, 40 byte packets
 1  172.21.128.2 (172.21.128.2)  5.759 ms  7.519 ms  6.094 ms  4.068 ms  3.982 ms
 2  10.20.30.2 (10.20.30.2)  4.951 ms  5.025 ms  4.749 ms  5.944 ms  4.309 ms
 3  212.175.32.141.static.ttnet.com.tr (212.175.32.141)  4.689 ms  4.405 ms  5.201 ms  4.802 ms  5.046 ms
 4  212.174.167.209 (212.174.167.209)  7.269 ms  6.771 ms  6.876 ms  7.115 ms  8.740 ms
 5  00-gayrettepe-sr14s-t2-1---00-gayrettepe-guvenlik-t3-1.statik.turktelekom.com.tr (81.212.221.114)  3511.071 ms  3504.358 ms  3520.910 ms  3410.842 ms  3443.494 ms
 6  * * * * *
 7  301-fra-col-2---00-ebgp-gayrettepe-sr12e-k.statik.turktelekom.com.tr (212.156.101.196)  40.037 ms  38.921 ms  38.747 ms  40.005 ms  40.872 ms
 8  * * 212.222.126.8 (212.222.126.8)  46.078 ms  43.902 ms  43.462 ms
 9  ae16.cr0-mtl1.ip4.gtt.net (89.149.186.134)  135.847 ms  222.083 ms  135.177 ms  134.724 ms  134.731 ms
10  ip4.gtt.net (208.116.159.206)  141.236 ms  140.690 ms  141.138 ms  141.846 ms *
11  imtrl-rq-ic-dmtrl-rq.risq.net (192.77.63.70)  275.006 ms  182.885 ms  321.995 ms  268.571 ms *
12  * * * * *
13  mcgill-gw.risq.net (132.202.62.90)  155.164 ms  154.718 ms  173.612 ms  173.864 ms  160.711 ms
14  * * * * *
```

Figure 4: Traceroute output with probe = 5

```
(base) zulaldemirtas@Zulals-MacBook-Pro-2 ~ % traceroute -q 10 www.mcgill.ca

traceroute to webcache.it.mcgill.ca (132.216.98.100), 64 hops max, 40 byte packets
 1  172.21.128.2 (172.21.128.2)  7.831 ms  4.224 ms  4.069 ms  3.843 ms  4.153 ms  3.814 ms *  4.233 ms  3.898 ms  6.764 ms
 2  10.20.30.2 (10.20.30.2)  4.177 ms  5.044 ms  5.423 ms  8.597 ms  5.055 ms  3.962 ms *  5.091 ms  4.731 ms  4.069 ms
 3  212.175.32.141.static.ttnet.com.tr (212.175.32.141)  5.010 ms  6.954 ms  5.675 ms  5.037 ms  5.945 ms  8.613 ms  5.051 ms  5.452 ms  5.612 ms  4.759 ms
 4  212.174.167.209 (212.174.167.209)  7.384 ms  6.157 ms  6.816 ms  6.954 ms  6.802 ms  7.048 ms  6.901 ms  10.677 ms  7.364 ms  6.775 ms
 5  00-gayrettepe-sr14s-t2-1---00-gayrettepe-guvenlik-t3-1.statik.turktelekom.com.tr (81.212.221.114)  3520.183 ms  3341.570 ms *  3346.625 ms  3494.173 ms  3518.613 ms  3497.095 ms  3483.409 ms  3425.590 ms  3275
.659 ms
 6  * * * * * * * * * *
 7  301-fra-col-2---00-ebgp-gayrettepe-sr12e-k.statik.turktelekom.com.tr (212.156.101.196)  42.859 ms  42.772 ms  48.118 ms  39.971 ms  45.565 ms  41.314 ms  41.178 ms  42.070 ms  40.060 ms  41.618 ms
 8  212.222.126.8 (212.222.126.8)  50.691 ms  43.321 ms  44.358 ms  43.617 ms  44.038 ms  44.986 ms  44.793 ms  44.139 ms  46.327 ms *
 9  ae16.cr0-mtl1.ip4.gtt.net (89.149.186.134)  134.734 ms  135.017 ms  296.982 ms  136.357 ms  134.902 ms  136.146 ms  137.828 ms  148.281 ms  135.737 ms  135.607 ms
10  ip4.gtt.net (208.116.159.206)  140.265 ms  140.996 ms  142.251 ms  142.510 ms *  142.856 ms  140.787 ms  182.658 ms *  147.730 ms
11  imtrl-rq-ic-dmtrl-rq.risq.net (192.77.63.70)  154.878 ms  155.787 ms  155.411 ms  155.443 ms *  155.526 ms  154.916 ms  155.755 ms  154.887 ms *
12  * * * * * * * * * *
13  mcgill-gw.risq.net (132.202.62.90)  154.635 ms  154.822 ms  154.075 ms  154.153 ms  154.644 ms *  155.131 ms  160.312 ms  155.556 ms  153.628 ms
14  * * * * * * * * * *
```

Figure 5: Traceroute output with probe = 10

5. What would happen if traceroute is used to find the path to an unassigned address? Does it matter if the network portion or only the host portion is unassigned? No screenshot is expected.

If traceroute is used to find the path to an unassigned address, the packets may still travel through the network until they reach a router that cannot forward them anymore. In this case, traceroute may show timeouts or ICMP unreachable messages and then stop. It matters whether the network portion or only the host portion is unassigned. If the network portion is unassigned, routers do not have a route to that network, so traceroute fails early in the path. If only the host portion is unassigned, the packets can still reach the destination network. However, the final host does not respond, so traceroute would fail at the last hop.

Part 2. Simulations with Cisco Packet Tracer

1. Attach a screenshot of the designed network with a label beside each port indicating its IP address, subnet mask, and gateway (if exists).

Figure 6 below shows the designed network. The network design consists of three branches: North, South, and Central. The CentralRouter is connected to the EdgeRouter, which is connected to the Internet Service Provider. Router-to-router WAN links use /30 subnets, while branch LANs use /23 subnetting. For the WAN links, I initially tried /31 subnetting using 2911, ISR4321, ISR4331, and PT8200 routers; however, it did not work in Cisco Packet Tracer. Therefore, /30 subnetting was used instead.

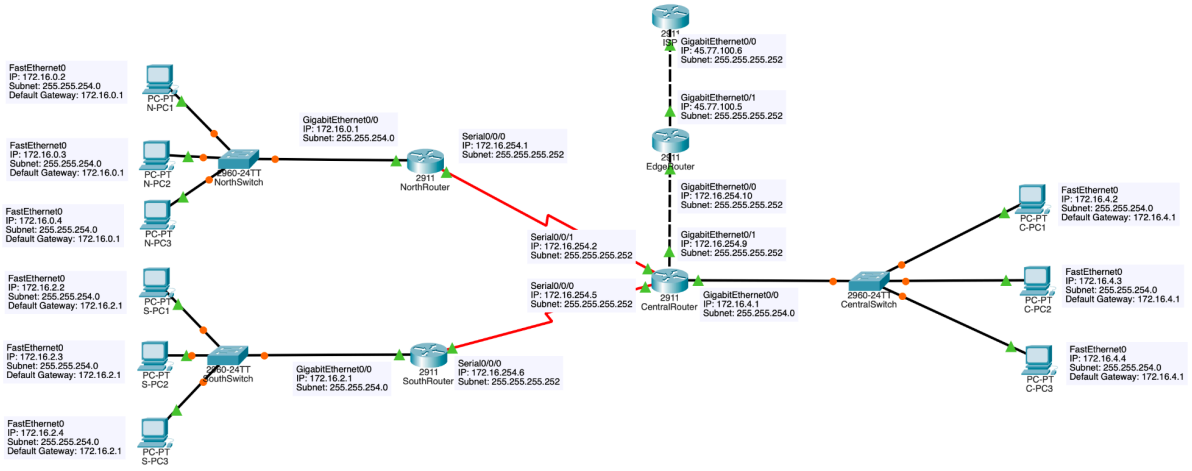


Figure 6: Network Design in Cisco Packet Tracer

2. How many branches does your network architecture support? What is the maximum number of devices that could be connected in each branch? Provide explanations. Show and explain your subnetting calculations (images of handwritten calculations are acceptable).

According to the calculations in Figure 7, a maximum of 510 devices can be connected in each branch, and the IP ranges allocated to each branch are shown. Since there are 2^{16} addresses in total in this private address block, $2^{16} / 2^{9} = 2^{7} = 128$ branches are supported in the current network architecture.

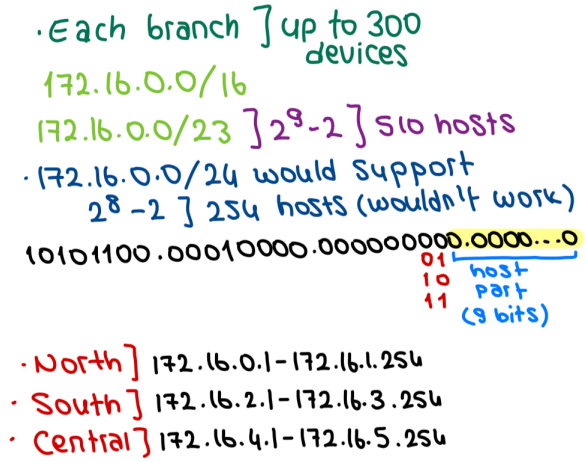


Figure 7: Subnetting Calculations

3. While configuring your network, you probably configured the routers using the Graphical User Interface. However, in real life, a command line is probably used. Make a table having two columns. The first column is the process, and the second one is the command. You can notice the commands written automatically when selecting an option in the GUI.

The configuration for the NorthRouter is as follows:

Process	**Command**
Select the LAN interface of the router	Router(config)#interface GigabitEthernet0/0
Configure the IPv4 address and Subnet Mask	Router(config-if)#ip address 172.16.0.1 255.255.254.0
Enable the interface	Router(config-if)#no shutdown

4. Find the CLI command that displays the routing table on a Cisco router. Attach a screenshot showing the routing table from one branch router. Briefly explain how the router knows how to reach the other branches.

The command that displays the routing table is *show ip route.* The routing table shows both directly connected networks and routes learned dynamically through the Routing Information Protocol (RIP). RIP advertises network information between routers so that each router learns how to reach the other branches via its neighboring routers. It uses the Bellman-Ford algorithm and hop count to calculate the best paths.

```
NorthRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.254.2 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.16.0.0/23 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
R       172.16.2.0/23 [120/2] via 172.16.254.2, 00:00:05, Serial0/0/0
R       172.16.4.0/23 [120/1] via 172.16.254.2, 00:00:05, Serial0/0/0
C       172.16.254.0/30 is directly connected, Serial0/0/0
L       172.16.254.1/32 is directly connected, Serial0/0/0
R       172.16.254.4/30 [120/1] via 172.16.254.2, 00:00:05, Serial0/0/0
R       172.16.254.8/30 [120/1] via 172.16.254.2, 00:00:05, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 172.16.254.2
```

Figure 8: Routing Table of NorthRouter

5. Your network should be working, it means any two devices can communicate. Attach screenshots of using the ping command to test connectivity between a device from each branch with a device from other branches.

Figures 8, 9, and 10 show the output of ping commands sent from devices located in different branches. It proves that two devices in different LANs can communicate with each other across the network.

```
C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.16.2.2: bytes=32 time=48ms TTL=125
Reply from 172.16.2.2: bytes=32 time=2ms TTL=125
Reply from 172.16.2.2: bytes=32 time=2ms TTL=125

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 48ms, Average = 17ms
```

Figure 9: ping from N-PC1 to S-PC1

```
C:\>ping 172.16.4.2

Pinging 172.16.4.2 with 32 bytes of data:

Reply from 172.16.4.2: bytes=32 time=21ms TTL=126
Reply from 172.16.4.2: bytes=32 time=2ms TTL=126
Reply from 172.16.4.2: bytes=32 time=48ms TTL=126
Reply from 172.16.4.2: bytes=32 time=18ms TTL=126

Ping statistics for 172.16.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 48ms, Average = 22ms
```

Figure 10: ping from S-PC1 to C-PC1

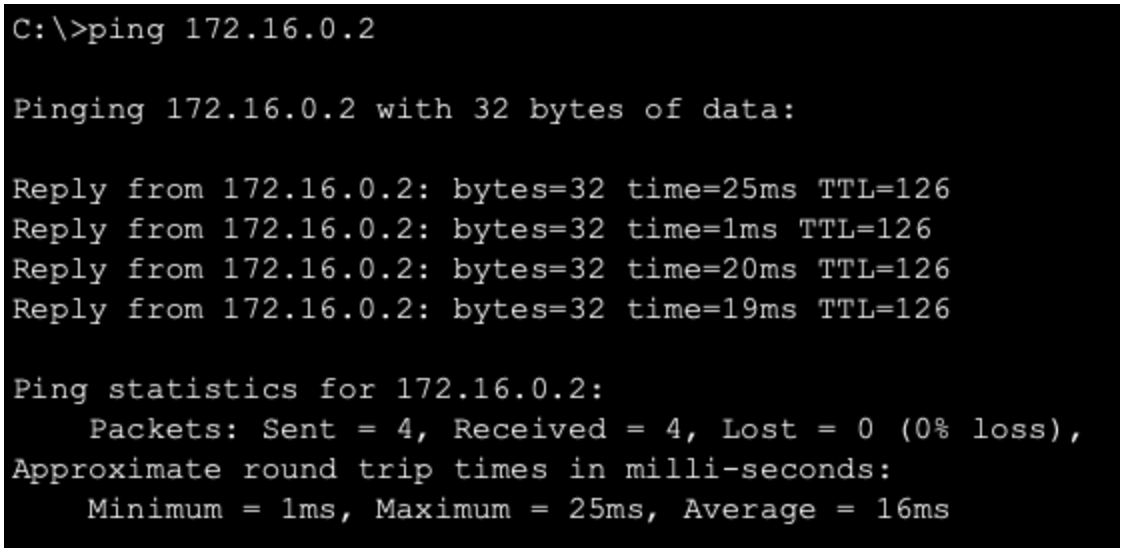


Figure 11: ping from C-PC1 to N-PC1

6. Use the CLI on each router to display the NAT translation table. Attach screenshots of the NAT table from each router.

- **What does a NAT translation entry represent?**
- **What do you notice about that NAT table? Does the table represent NAT or Port Address Translation (PAT)? Why?**

The command to display the NAT table is *show ip nat translations.* In this network, only the EdgeRouter has a NAT table because it is the only router connected to the external network. The other routers do not show any NAT entries since they do not perform NAT. A NAT translation entry represents the mapping between a private internal IP address and the public IP address. It allows internal networks to communicate with the internet.

From Figure 12, it can be seen that multiple IP addresses are translated to the same public IP address using different identifiers. This shows the network uses PAT since it allows multiple internal IP addresses to share the same public IP address. Since this network design only has one public IP address, PAT is required.

```
EdgeRouter>show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 45.77.100.5:1     172.16.0.2:1       45.77.100.6:1      45.77.100.6:1
icmp 45.77.100.5:2     172.16.0.2:2       45.77.100.6:2      45.77.100.6:2
icmp 45.77.100.5:3     172.16.0.2:3       45.77.100.6:3      45.77.100.6:3
icmp 45.77.100.5:4     172.16.0.2:4       45.77.100.6:4      45.77.100.6:4
```

Figure 12: NAT Translation table on EdgeRouter

```
NorthRouter>enable
NorthRouter#show ip nat translations
NorthRouter#
```

Figure 13: NAT Translation table on NorthRouter

```
SouthRouter>enable
SouthRouter#show ip nat translations
SouthRouter#
```

Figure 14: NAT Translation table on SouthRouter

```
CentralRouter>enable
CentralRouter#show ip nat translations
CentralRouter#
```

Figure 15: NAT Translation table on CentralRouter

References

[1] “Traceroute(8) — Linux manual page,” *Linux Die*. [Online]. Available: https://linux.die.net/man/8/traceroute

[2] “Private network,” *Wikipedia*. [Online]. Available: https://en.wikipedia.org/wiki/Private_network